

Untyped Lambda Calculus

Chris Anto Fröschl

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Chapter 1

Alpha equivalence equalities

Definition 1 (Agreement set). For permutations π, π' of the set of atoms $\mathbb{A}$, the *agreement set* is

$$AS(\pi, \pi') \stackrel{\text{def}}{=} \{ a \in \mathbb{A} \mid \pi(a) = \pi'(a) \}.$$

([Crole2012], Section 2.)

Definition 2 (Disagreement set). For permutations π, π' of the set of atoms $\mathbb{A}$, the *disagreement set* is

$$DS(\pi, \pi') \stackrel{\text{def}}{=} \{ a \in \mathbb{A} \mid \pi(a) \neq \pi'(a) \}.$$

([Crole2012], Section 2.)

Definition 3 (Permutation action on expressions). The action $\pi \cdot E$ of a (finitely supported) permutation π of atoms on an expression E is defined by recursion on E :

$$\pi \cdot a \stackrel{\text{def}}{=} \pi(a), \quad \pi \cdot P(E_1, E_2) \stackrel{\text{def}}{=} P(\pi \cdot E_1, \pi \cdot E_2), \quad \pi \cdot B([a]E) \stackrel{\text{def}}{=} B([\pi \cdot a] \pi \cdot E).$$

([Crole2012], Section 2.)

Definition 4 (Atom swapping). For atoms x, y , the transposition (xy) is the permutation exchanging x and y and fixing every other atom; its action $(xy) \cdot E$ on an expression E is the special case of theorem 3 in which π is a transposition, so that $(xy) \cdot E$ is E with all occurrences of x and y swapped. ([Crole2012], Section 2; [Gabbay2002], Section 2.)

Definition 5 (Definition 3.2: the relation $\sim_{p\#}$). The binary relation $\sim_{p\#}$ on the set of expressions is inductively defined by

$$\frac{}{a \sim_{p\#} a} \text{atom} \quad \frac{E_1 \sim_{p\#} E'_1 \quad E_2 \sim_{p\#} E'_2}{P(E_1, E_2) \sim_{p\#} P(E'_1, E'_2)} \text{pcg} \quad \frac{(za) \cdot E \sim_{p\#} (zb) \cdot E'}{B([a]E) \sim_{p\#} B([b]E')} \pi\# \quad [z \# a, b, E, E']$$

where the side condition $z \# a, b, E, E'$ means that z differs from a and b and has no free occurrence in E or E' . ([Crole2012], Definition 3.2.)

Definition 6 (Definition 3.3: the relation $\sim_p^1$). The binary relation $\sim_p^1$ on expressions is defined by the rules of ??, but with the rule π replaced by

$$\frac{(za) \cdot E \sim_p^1 (zb) \cdot E'}{B([a]E) \sim_p^1 B([b]E')} \pi_1 \quad [z \not\# E, E']$$

that is, the witness z is only required not to occur in the two bodies. ([Crole2012], Definition 3.3.)

Definition 7 (The relation $\sim_p^2$ of Proposition 4.3). The binary relation $\sim_p^2$ on expressions is defined by the rules of ??, but with the rule π replaced by

$$\frac{(za) \cdot E \sim_p^2 (zb) \cdot E'}{B([a]E) \sim_p^2 B([b]E')} \pi_2 \quad [z \not\vdash a, b]$$

that is, the witness z need only differ from the two binding atoms. ([Crole2012], Proposition 4.3.)

Definition 8 (The relation $\sim_p^3$ of Proposition 4.3). The binary relation $\sim_p^3$ on expressions is defined by the rules of ??, but with the rule π replaced by

$$\frac{(za) \cdot E \sim_p^3 (zb) \cdot E'}{B([a]E) \sim_p^3 B([b]E')} \pi_3$$

which carries no side condition at all. ([Crole2012], Proposition 4.3.)

Definition 9 (Definition 3.4: the relation $\sim_r$). The binary relation $\sim_r$ on expressions is inductively defined by

$$\begin{array}{c} \overline{E \sim_r E} \text{ ref} \quad \frac{E_1 \sim_r E_2}{E_2 \sim_r E_1} \text{ sym} \quad \frac{E_1 \sim_r E_2 \quad E_2 \sim_r E_3}{E_1 \sim_r E_3} \text{ trs} \\ \\ \frac{E_1 \sim_r E'_1 \quad E_2 \sim_r E'_2}{P(E_1, E_2) \sim_r P(E'_1, E'_2)} \text{ pcg} \quad \frac{E \sim_r E'}{B([a]E) \sim_r B([a]E')} \text{ bcg} \\ \\ \overline{B([a]E) \sim_r B([a']E\{a'/a\})} \alpha \quad [a' \not\vdash a, E] \end{array}$$

where $E\{a'/a\}$ is capture-avoiding substitution and the side condition $a' \not\vdash a, E$ says that a' differs from a and does not occur at all in E . ([Crole2012], Definition 3.4.)

Definition 10 (Definition 3.5: the relation $\sim_{r\#}$). The binary relation $\sim_{r\#}$ on expressions is defined by the rules of theorem 9, but with the rule α replaced by

$$\overline{B([a]E) \sim_{r\#} B([a']E\{a'/a\})} \alpha\# \quad [a' \# a, E]$$

whose side condition only requires a' to differ from a and to have no *free* occurrence in E . ([Crole2012], Definition 3.5.)

Lemma 11 (Lemma 6.1: transpositions preserve $\sim_p$). *For any atoms u and v and expressions E and E' we have*

$$E \sim_p E' \implies (uv) \cdot E \sim_p (uv) \cdot E'.$$

([Crole2012], Lemma 6.1.)

Proof. □

Lemma 12 (Lemma 6.7: substitution preserves α -equivalence). *For any expressions E, E', R, R' and any atom a ,*

$$E \sim_p E' \text{ and } R \sim_p R' \implies E\{R/a\} \sim_p E'\{R'/a\}.$$

This is Lemma 6.7 of [Crole2012], which is stated there for $\sim_r$ and for the renaming $E\{a/b\}$; by Theorem 4.4 the two statements agree, and the version formalised here allows an arbitrary expression to be substituted, and its two sides to differ up to α -equivalence.

Proof. □

1.1 Lemmas on expressions

The lemmas of Section 6.1 of [Crole2012], which support the proofs of the equalities of Section 4.

Lemma 13 (Lemma 6.2(1): permutations agreeing on occurring atoms). *For any expression E and permutations π, π' ,*

$$\text{occ}(E) \subseteq AS(\pi, \pi') \implies \pi \cdot E = \pi' \cdot E.$$

([Crole2012], Lemma 6.2 part 1.)

Proof. □

Lemma 14 (Lemma 6.2(1) for transpositions). *For any expression E and atoms a, u, z with $u \not\vdash E$ and $z \not\vdash E$,*

$$(zu) \cdot (ua) \cdot E = (za) \cdot E.$$

This is the instance of theorem 13 used in [Crole2012] in the proofs of Theorems 4.2 and 4.4.

Proof. □

Lemma 15 (Lemma 6.2(2): permutations agreeing on free atoms). *For any expression E and permutations π, π' ,*

$$\text{free}(E) \subseteq AS(\pi, \pi') \implies \pi \cdot E \sim_p \pi' \cdot E.$$

([Crole2012], Lemma 6.2 part 2.)

Proof. □

Lemma 16 (Lemma 6.2(2) for transpositions). *For any expression E and atoms a, u, z with $u \# E$ and $z \# E$,*

$$(zu) \cdot (ua) \cdot E \sim_p (za) \cdot E.$$

This is the instance of theorem 15 used in [Crole2012] in the proof of Theorem 4.1.

Proof. □

Lemma 17 (Lemma 6.3, in naive-renaming form). *For any atoms a, z and any expression E with $z \not\vdash E$,*

$$E[a \mapsto z] \sim_r E\{z/a\}$$

where $E[a \mapsto z]$ is the naive renaming of $??$. This is the form in which the induction of Lemma 6.3 of [Crole2012] is carried out here.

Proof. □

Lemma 18 (Lemma 6.3: swapping versus substitution, for $\sim_r$). *For any atoms a and z and any expression E ,*

$$z \not\vdash E \implies (za) \cdot E \sim_r E\{z/a\}.$$

([Crole2012], Lemma 6.3.)

Proof. □

Lemma 19 (Lemma 6.4, in naive-renaming form). *For any atoms a, z and any expression E with $z \not\vdash E$,*

$$E[a \mapsto z] \sim_{r\#} E\{z/a\}.$$

This is theorem 17 with every use of the rule α replaced by $\alpha\#$.

Proof. □

Lemma 20 (Lemma 6.4: swapping versus substitution, for $\sim_{r\#}$). *For any atoms a and z and any expression E ,*

$$z \not\vdash E \implies (za) \cdot E \sim_{r\#} E\{z/a\}.$$

([Crole2012], Lemma 6.4.)

Proof. □

Lemma 21 (Lemma 6.6: avoiding an atom, for $\sim_r$). *For all expressions E and atoms a ,*

$$a \# E \implies (\exists \hat{E}) (a \not\vdash \hat{E} \wedge \hat{E} \sim_r E).$$

Informally, bound atoms can always be renamed so that a particular atom does not occur; this may be regarded as a form of Barendregt's variable convention. ([Crole2012], Lemma 6.6.)

Proof. □

Lemma 22 (Lemma 6.6 for $\sim_p$). *For all expressions E and atoms a ,*

$$a \# E \implies (\exists \hat{E}) (a \not\vdash \hat{E} \wedge \hat{E} \sim_p E).$$

This is theorem 21 with $\sim_r$ replaced by $\sim_p$. It cannot be obtained from theorem 21 through Theorem 4.4, because the proof of that theorem uses this statement.

Proof. □

Lemma 23 (Lemma 6.3 for $\sim_p$, in naive-renaming form). *For any atoms a, a' and any expression E with $a' \not\vdash E$,*

$$E[a \mapsto a'] \sim_p E\{a'/a\}.$$

This is theorem 17 with $\sim_r$ replaced by $\sim_p$.

Proof. □

Lemma 24 (Substituting a fresh atom is a transposition). *For any expression E and atoms a, a' with $a' \# E$,*

$$E\{a'/a\} \sim_p (a a') \cdot E.$$

This is the first of the two ingredients of theorem 29: note that a' is only required to be fresh for E , bound occurrences being renamed away first using theorem 22.

Proof. □

Lemma 25 (Splitting a transposition through a fresh atom). *For any expression E and atoms z, a, a' with $z \# E$ and $a' \# E$,*

$$(za) \cdot E \sim_p (za') \cdot (a a') \cdot E.$$

This is the second ingredient of theorem 29: the two permutations agree on $\text{free}(E)$, so theorem 15 applies.

Proof. □

Definition 26 (Chain of transpositions). Given a finite list of triples of atoms $(z_1, a_1, a'_1), \dots, (z_n, a_n, a'_n)$ and an expression E , the expression

$$(z_1 a_1) \cdot \dots \cdot (z_n a_n) \cdot E$$

which is the left-hand side of theorem 29.

Definition 27 (Chain of transpositions after a chain of substitutions). Given a finite list of triples of atoms $(z_1, a_1, a'_1), \dots, (z_n, a_n, a'_n)$ and an expression E , the expression

$$(z_1 a'_1) \cdot \dots \cdot (z_n a'_n) \cdot E\{a'_1/a_1\} \dots \{a'_n/a_n\}$$

which is the right-hand side of theorem 29.

Definition 28 (Atoms of a chain). The list of all atoms z_i, a_i, a'_i mentioned by a list of triples $(z_1, a_1, a'_1), \dots, (z_n, a_n, a'_n)$. Its being repetition-free is the pairwise-disjointness hypothesis of theorem 29.

Lemma 29 (Lemma 6.5: chains of transpositions versus chains of substitutions). *Let the atoms z_i, a_i, a'_i be pairwise disjoint across all i . Then for every expression E with $z_i \# E$ and $a'_i \# E$ for all i ,*

$$(z_1 a_1) \cdot \dots \cdot (z_n a_n) \cdot E \sim_p (z_1 a'_1) \cdot \dots \cdot (z_n a'_n) \cdot E\{a'_1/a_1\} \dots \{a'_n/a_n\}.$$

([Crole2012], Lemma 6.5.) *The hypotheses $z_i \nvdash E$ of the paper are weakened here to $z_i \# E$, which suffices because both ingredients of the proof are invariant under $\sim_p$ and mention only free atoms.*

Proof. □

Lemma 30 (Lemma 6.5 for a single triple of atoms). *For any expression E and atoms z, a, a' with $z \nvdash E$, $z \neq a$, $a \neq a'$ and $a' \# E$,*

$$(z a) \cdot E \sim_p (z a') \cdot E\{a'/a\}.$$

This is the case $n = 1$ of theorem 29, which is the step carried out in [Crole2012] inside the proof of Theorem 4.4 for the rule α .

Proof. □

1.2 The equality of the definitions

The five relations of Section 3 of [Crole2012] are all the same relation, each capturing α -equivalence for expressions.

Lemma 31 ($\sim_p$ is contained in $\sim_{p\#}$). *For all expressions E, E' we have $E \sim_p E' \implies E \sim_{p\#} E'$. This is the trivial half of Theorem 4.1 of [Crole2012]: non-occurrence implies freshness.*

Proof. □

Lemma 32 ($\sim_{p\#}$ is contained in $\sim_p$). *For all expressions E, E' we have $E \sim_{p\#} E' \implies E \sim_p E'$. Equivalently, $\sim_p$ is closed under the rules for $\sim_{p\#}$; the only interesting case is the rule $\pi\#$, which is handled by picking a fully non-occurring witness and appealing to theorem 11 and theorem 15. ([Crole2012], Theorem 4.1.)*

Proof. □

Theorem 33 (Theorem 4.1: $\sim_p$ equals $\sim_{p\#}$). *The relation $\sim_p$ of α -equivalence according to ?? is identical to the relation $\sim_{p\#}$ of theorem 5. ([Crole2012], Theorem 4.1.)*

Proof. □

Lemma 34 ($\sim_p$ is contained in $\sim_p^1$). *For all expressions E, E' we have $E \sim_p E' \implies E \sim_p^1 E'$, since the side condition of π is stronger than that of π_1 . ([Crole2012], Theorem 4.2.)*

Proof. □

Lemma 35 (The degenerate case of the rule π_1). *For atoms a, b and expressions E, E' with $a \not\vdash E'$,*

$$E \sim_p (a b) \cdot E' \implies B([a]E) \sim_p B([b]E').$$

This is the case $u = a$ in the proof of Theorem 4.2 of [Crole2012], which is settled using theorem 11 and theorem 13.

Proof. □

Lemma 36 ($\sim_p^1$ is contained in $\sim_p$). *For all expressions E, E' we have $E \sim_p^1 E' \implies E \sim_p E'$. Equivalently, $\sim_p$ is closed under the rules for $\sim_p^1$: if the witness u of the rule π_1 differs from both binding atoms one appeals directly to π , and otherwise one may assume without loss of generality that $u = a$ and use theorem 35. ([Crole2012], Theorem 4.2.)*

Proof. □

Theorem 37 (Theorem 4.2: $\sim_p$ equals $\sim_p^1$). *The relation $\sim_p$ of α -equivalence according to ?? is identical to the relation $\sim_p^1$ of theorem 6. ([Crole2012], Theorem 4.2.)*

Proof. □

Proposition 38 (Proposition 4.3: variants of the side condition of π). *Consider the variants $\sim_p^i$ of ?? obtained by altering the side condition of the rule π :*

rule	relation	condition	$= \sim_p$
π	$\sim_p$	$z \not\vdash a, b, E, E'$	yes
π_1	$\sim_p^1$	$z \not\vdash E, E'$	yes
π_2	$\sim_p^2$	$z \not\vdash a, b$	no
π_3	$\sim_p^3$	none	no

The last column states for which i the relation $\sim_p^i$ is equal to $\sim_p$: the first two rows hold by theorem 37, while for $a \neq b$ the pairs $B([a]P(a, z)), B([b]P(b, a))$ and $B([a]P(b, a)), B([b]P(a, b))$ are respectively related by $\sim_p^2$ and by $\sim_p^3$ but are not α -equivalent, having different free atoms. ([Crole2012], Proposition 4.3.)

Proof. □

Lemma 39 ($\sim_r$ is contained in $\sim_p$). *For all expressions E, E' we have $E \sim_r E' \implies E \sim_p E'$. Equivalently, $\sim_p$ is closed under the rules for $\sim_r$: the rules ref, sym, trs and bcg are ??, ??, ?? and ??, pcg is immediate, and the rule α is the instance theorem 30 of Lemma 6.5. ([Crole2012], Theorem 4.4.)*

Proof. □

Lemma 40 ($\sim_p$ is contained in $\sim_r$). For all expressions E, E' we have $E \sim_p E' \implies E \sim_r E'$. Equivalently, $\sim_r$ is closed under the rules for $\sim_p$; for the rule π one appeals to theorem 18 twice, to relate $(za) \cdot E$ with $E\{z/a\}$ and $(zb) \cdot E'$ with $E'\{z/b\}$, and then uses bcg, two instances of α , sym and trs. ([Crole2012], Theorem 4.4.)

Proof. □

Theorem 41 (Theorem 4.4: $\sim_p$ equals $\sim_r$). The relation $\sim_p$ of α -equivalence according to ?? is identical to the relation $\sim_r$ of theorem 9. ([Crole2012], Theorem 4.4.)

Proof. □

Lemma 42 ($\sim_{r\#}$ is contained in $\sim_p$). For all expressions E, E' we have $E \sim_{r\#} E' \implies E \sim_p E'$. The cases are exactly those of theorem 39, with the rule α replaced by $\alpha\#$; the latter is where the strengthening theorem 29 of the paper's inductive argument is needed, here in the form theorem 30. ([Crole2012], Theorem 4.5.)

Proof. □

Lemma 43 ($\sim_p$ is contained in $\sim_{r\#}$). For all expressions E, E' we have $E \sim_p E' \implies E \sim_{r\#} E'$. The steps are exactly those of theorem 40, with theorem 20 used in place of theorem 18 and the two instances of α replaced by $\alpha\#$, which is legitimate because $z \not\vdash E, E'$ implies $z \# E, E'$. ([Crole2012], Theorem 4.5.)

Proof. □

Theorem 44 (Theorem 4.5: $\sim_p$ equals $\sim_{r\#}$). The relation $\sim_p$ of α -equivalence according to ?? is identical to the relation $\sim_{r\#}$ of theorem 10. ([Crole2012], Theorem 4.5.)

Proof. □

Theorem 45 (Theorem 4.6: $\sim_r$ equals $\sim_{r\#}$). The relation $\sim_r$ of α -equivalence according to theorem 9 is identical to the relation $\sim_{r\#}$ of theorem 10. ([Crole2012], Theorem 4.6.)

Proof. Immediate from theorem 41 and theorem 44. □